# Enterprise Systems Architecture Report

## Executive Overview and Scope

This document establishes the production architectural standards for the next-generation distributed transaction processing system.

## 1. High-Level System Objectives

All core subsystems are designed for fault-tolerant operation across three geographic regions with active-active routing topologies.

1. Achieve 99.99% availability across multi-region deployment zones.
2. Maintain p99 API response latencies below 85 milliseconds under peak loads.
3. Enforce zero-trust mutual TLS authentication between all internal services.
4. Guarantee automated database failover within 120 seconds of telemetry loss.

## 2. Core Service Infrastructure

The microservice plane isolates stateless business logic from stateful storage engines, facilitating elastic horizontal autoscaling during unpredictable traffic surges.

| Component | Technology | Target Latency | Redundancy |
|---|---|---|---|
| API Gateway | Envoy Proxy | < 5 ms | N+2 Active |
| Auth Engine | OAuth2 / OIDC | < 12 ms | Multi-Region |
| Event Broker | Apache Kafka | < 15 ms | 3x Replicas |
| Primary Store | PostgreSQL HA | < 25 ms | Synchronous Leader |

### 2.1 Event Streaming and Messaging

Event-driven streaming via distributed commit logs enables decoupled communication patterns and reliable change data capture between asynchronous consumers.

## 3. Security, Compliance, and Data Governance

Enterprise compliance requirements necessitate immutable audit logging and strict cryptographic isolation for customer financial transactions.

- Hardware Security Module key rotation enforced every 90 days
- Role-based access control policies validated via automated CI tests
- Continuous static and dynamic vulnerability analysis on all container images

## 4. Architectural Conclusions and Next Steps

Implementation will proceed in staged delivery milestones, commencing with canary verification of edge gateway proxies and foundational database replication.